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August 31, 1967
SUPERSEDING
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May 13, 1964

FEDERAL SPECIFICATION

DRILL, MASONRY, ROTARY DRIVEN (CORELESS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the coreless type of carbide tipped drill used for drilling holes in concrete and other masonry materials.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Type and sizes. The drills shall be of one type and of the sizes specified (see table I and 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Standards:
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, Los Angeles, and Auburn, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:
MIL-H-15424 - Hand Tools; Packaging of.

FSC 5130

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Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustration. The illustration shown herein is for the convenience of identification and is not intended to preclude the purchase of drills which are otherwise in accordance with this specification.

3.2 Material. The drill body and shank shall be steel, and the cutting tip shall be sintered carbide.

3.3 Hardness. The hardness of the body and shank shall be in accordance with the manufacturer's standard practice. The carbide tip hardness shall be not less than Rockwell A 90.5.

3.4 Design. Drills shall be designed for right-hand cutting. The drill shall have a carbide tip securely attached to one end of a fluted body and a straight shank on the other end. The type and shape of the flutes and the relief angles on the carbide tip shall be in accordance with the manufacturer's standard practice. Drills shall be similar to figure 1, and meet the tests in section 4.

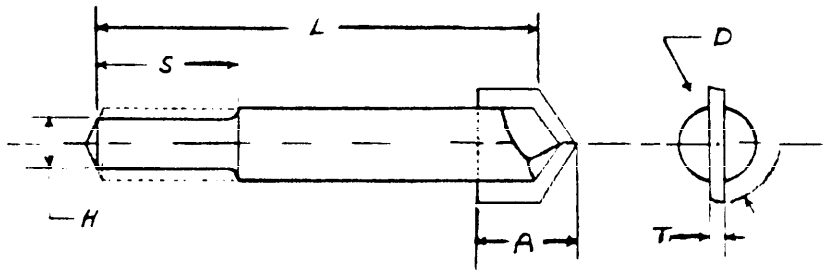



FIGURE 1. Drill, masonry, rotary driven.

3.5 Sizes and dimensions. Sizes and dimensions shall be in accordance with table I, and as specified in 6.1.



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TABLE I. Drill sizes and dimensions

| Drill designation | Nominal drill diameter | Actual diameter D ± 0.000 - .010 | Tip thickness T min. | Tip length L min. | Overall length ± 1/8 | Shank | |
|---|---|---|---|---|---|---|---|
| | | | | | | Length S min. | Diameter H ± 0.015 - .005 |
| | Inches | Inches | Inches | Inches | Inches | Inches | Inches |
| FMD 3 | 3/16 | 0.208 | 0.025 | 0.175 | 3 | - - | 11/64 |
| FMD 4 | 1/4 | .270 | .040 | .220 | 4 | - - | 15/64 |
| FMD 5 | 5/16 | .338 | .050 | .310 | 4 | 1-1/4 | 1/4 |
| FMD 6 | 3/8 | .400 | .050 | .368 | 4 | 1-1/4 | 1/4 |
| FMD 7 | 7/16 | .468 | .050 | .468 | 6 | 1-3/8 | 1/4 |
| FMD 7-1 | 7/16 | .468 | .050 | .468 | 6 | 1-3/8 | 3/8 |
| FMD 8 | 1/2 | .535 | .081 | .472 | 6 | 1-3/8 | 3/8 |
| FMD 8-1 | 1/2 | .535 | .081 | .472 | 6 | 1-3/8 | 1/4 |
| FMD 9 | 9/16 | .597 | .081 | .472 | 6 | 1-3/8 | 1/2 |
| FMD 9-1 | 9/16 | .597 | .081 | .472 | 6 | 1-3/8 | 7/16 |
| FMD 10 | 5/8 | .660 | .081 | .598 | 6 | 1-3/8 | 1/2 |
| FMD 11 | 11/16 | .723 | .081 | .598 | 6 | 1-3/8 | 1/2 |
| FMD 12 | 3/4 | .785 | .081 | .598 | 6 | 1-3/8 | 1/2 |
| FMD 14 | 7/8 | .915 | .081 | .728 | 6 | 1-3/8 | 1/2 |
| FMD 16 | 1 | 1.040 | .081 | .728 | 6 | 1-3/8 | 1/2 |
| FMD 18 | 1-1/8 | 1.170 | .081 | .858 | 6 | 1-5/8 | 1/2 |
| FMD 20 | 1-1/4 | 1.295 | .112 | .858 | 10 | 1-5/8 | 1/2 |
| FMD 24 | 1-1/2 | 1.545 | .112 | 1.045 | 10 | 1-5/8 | 1/2 |

3.6 Finish. Body of drill shall be chrome or cadmium plated, or otherwise treated to provide a smooth surface for removal of drilling debris, and provide a corrosion-resistant finish.

3.7 Identification marking. Each drill shall be marked in a permanent and legible manner with the nominal drill diameter and the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.8 Workmanship. Workmanship shall be first class in every respect. Drills shall be free from rust, laps, cracks, and other imperfections which may impair their serviceability or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table II.



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**TABLE II. Sampling data**

| Category | Sample unit | Inspection level | Acceptable quality level | AQL expressed in terms of | Reference |
|---|---|---|---|---|---|
| Visual examination | 1 each | II | 1.5 | Defects per hundred units | 4.3.1 |
| Dimensional examination | 1 each | S-4 | 1.5 | do. | 4.3.2 |
| Testing | 1 each | S-4 | 2.5 | do. | 4.4 |
| Preparation for delivery | One container | S-2 | 4.0 | do. | 4.5 |

4.3 **Examination.**

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking.

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 and 4.4.2.

4.4.1 Hardness test. Hardness test shall be conducted in accordance with method 241 of Fed. Test Method Std. No. 151.

4.4.2 Drilling test. Each sample drill shall be tested by drilling holes in standard mixed 3-2-1 concrete with sandstone aggregate (all aggregate less than 1/4 inch). The concrete shall be cured for at least 28 days prior to test. Ten holes shall be drilled 3 inches deep for drills 6 inches and longer overall length, and 20 holes shall be drilled 1-1/2 inches deep for shorter drills. The drilling shall be performed dry, without coolant. The drill shall be rejected if any permanent distortion or breakage occurs, or if the depth of penetration is less than one inch per minute. The holes shall be drilled without resharpening of the drills.

4.5 Inspection of preparation for delivery. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Unless otherwise specified (see 6.1), the drills shall be preserved, packaged, and packed in accordance with MIL-H-15464. The level of preservation and packaging shall be A or C, and the level of packing shall be A, B, or C, as specified (see 6.1).

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the unit packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to markings required by the contract or order, the unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size required (see table I).
(c) Selection of levels of preservation, packaging, and packing (see 5.1).

Military Interest:
DOD coordination has been waived.

Civil Interest:
GSA-FSS

Preparing Activity:
GSA-FSS

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Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price of this specification 5 cents each.



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